

Blockchain-Based Secure Electronic Health Record (EHR) Management System

1. Introduction

The Blockchain-Based Secure Electronic Health Record (EHR) Management System is a web-based healthcare application designed to securely manage patient medical records. It allows administrators to manage patients, doctors, and medical records while protecting sensitive information using blockchain technology. Each medical record is linked using SHA-256 cryptographic hashes, making unauthorized modifications easy to detect.

2. Software Used

Software	Purpose
Python 3.x	Backend
Flask	Web Framework
SQLite	Database Management
SQLAlchemy	Database ORM
HTML/CSS	Frontend
Blockchain	Secure record linking
SHA-256	Hashing Algorithm
JavaScript	Client-Side function

3. Project Modules

1. Login Module

- Administrator Login
- Secure authentication
- Dashboard access

2. Patient Management

- Register new patients
- View patient details
- Edit patient information
- Delete patient records

3. Doctor Management

- Add doctor details
- View doctor list
- Manage doctor information

4. Medical Record Management

- Add patient medical records
- Store diagnosis
- Store disease details
- Store prescriptions
- Record visit date

5. Blockchain Module

- Creates Genesis Block
- Generates SHA-256 hash
- Links each block using Previous Hash
- Stores Current Hash
- Maintains blockchain integrity

6. Blockchain Verification

- Checks blockchain validity
- Detects unauthorized modifications
- Confirms data integrity

7. Blockchain Explorer

Displays all blockchain blocks including:

- Block Number
- Timestamp
- Previous Hash
- Current Hash
- Medical Record Details

8. Dashboard

Displays

- Total Patients
- Total Doctors
- Total Medical Records
- Total Blockchain Blocks

4. Source Code:

4.1 Database Connection (app.py)

```
from flask import Flask
```

```
from models import db
```

```
app = Flask(__name__)
```

```
app.config["SQLALCHEMY_DATABASE_URI"] = "sqlite:///ehr.db"

app.config["SQLALCHEMY_TRACK_MODIFICATIONS"] = False

db.init_app(app)

with app.app_context():

    db.create_all()
```

4.2 PATIENT MODEL(models.py)

```
class Patient(db.Model):

    __tablename__ = "patients"

    id = db.Column(db.Integer, primary_key=True)

    patient_id = db.Column(db.String(20), unique=True)

    name = db.Column(db.String(100))

    age = db.Column(db.Integer)

    gender = db.Column(db.String(20))

    blood_group = db.Column(db.String(10))

    phone = db.Column(db.String(15))

    address = db.Column(db.String(250))
```

4.3 Doctor Model (models.py)

```
class Doctor(db.Model):

    __tablename__ = "doctors"

    id = db.Column(db.Integer, primary_key=True)

    doctor_id = db.Column(db.String(20), unique=True)

    name = db.Column(db.String(100))

    specialization = db.Column(db.String(100))

    email = db.Column(db.String(100))

    phone = db.Column(db.String(15))
```

4.4 Medical Record Model

```
class MedicalRecord(db.Model):

    __tablename__ = "medical_records"

    id = db.Column(db.Integer, primary_key=True)

    patient_id = db.Column(db.Integer,
                           db.ForeignKey("patients.id"))

    doctor_id = db.Column(db.Integer,
                           db.ForeignKey("doctors.id"))

    disease = db.Column(db.String(100))

    diagnosis = db.Column(db.String(300))

    prescription = db.Column(db.String(300))

    visit_date = db.Column(db.String(20))

    previous_hash = db.Column(db.String(64))

    current_hash = db.Column(db.String(64))
```

4.5 Blockchain Class

```
import hashlib

import json

from time import time

class Blockchain:

    def __init__(self):

        self.chain = []

        self.create_block("0", "Genesis Block")
```

4.6 Creating a Blockchain Block

```
def create_block(self, previous_hash, data):

    block = {

        "index": len(self.chain) + 1,

        "timestamp": str(time()),
```

```

        "data": data,

        "previous_hash": previous_hash
    }

    block["hash"] = self.hash(block)

    self.chain.append(block)

    return block

```

4.7 SHA-256 Hash Generation

```

def hash(self, block):

    encoded = json.dumps(

        block,

        sort_keys=True

    ).encode()

    return hashlib.sha256(encoded).hexdigest()

```

4.8 Adding a Medical Record

```

record = MedicalRecord(

    patient_id=int(request.form["patient"]),

    doctor_id=int(request.form["doctor"]),

    disease=request.form["disease"],

    diagnosis=request.form["diagnosis"],

    prescription=request.form["prescription"],

    visit_date=datetime.now().strftime("%d-%m-%Y"),

    previous_hash=previous_hash,

    current_hash=block["hash"]

)

db.session.add(record)

db.session.commit()

```

4.9 Blockchain Verification

```
@app.route("/verify_blockchain")

def verify_blockchain():

    if blockchain.is_chain_valid():

        return "Blockchain is Valid"

    else:

        return "Blockchain has been Tampered"
```

4.10 Dashboard Statistics

```
@app.route("/dashboard")

def dashboard():

    total_patients = Patient.query.count()

    total_doctors = Doctor.query.count()

    total_records = MedicalRecord.query.count()

    total_blocks = MedicalRecord.query.count()

    return render_template(

        "dashboard.html",

        total_patients=total_patients,

        total_doctors=total_doctors,

        total_records=total_records,

        total_blocks=total_blocks

    )
```

5.OUTPUT SCREENSHOTS

6.CONCLUSION

The Blockchain-Based Secure Electronic Health Record (EHR) Management System demonstrates how blockchain technology can improve the security and reliability of healthcare data. By linking each medical record with cryptographic hashes, the system ensures that records remain authentic and tamper-evident.

The project combines Flask, SQLite, SQLAlchemy, and Blockchain to provide a secure and user-friendly platform for managing patients, doctors, and medical records. It highlights the practical application of blockchain in healthcare and shows how modern technologies can be used to protect sensitive patient information while maintaining transparency and trust.

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